

Bias in Generative AI

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Informatics

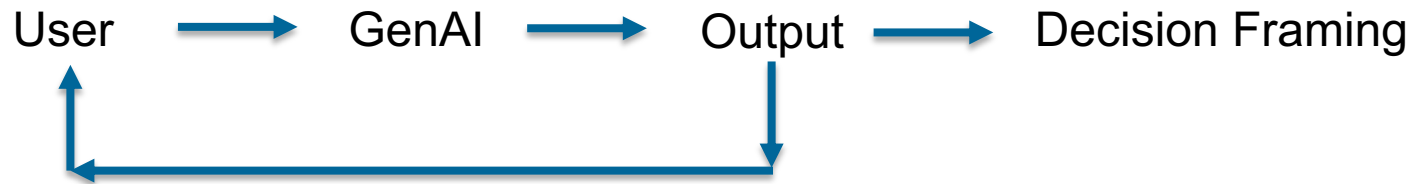


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Generative AI in Decision Contexts

- Increasing use of GenAI as an interface to information
- Applied in relocation, hiring, education, healthcare, and business support
- Outputs influence framing, salience, and perceived options
- Shift from retrieval-based to generative systems



[Ferrara, 2024; Dudy et al., 2025]



Conceptualizing Bias in GenAI

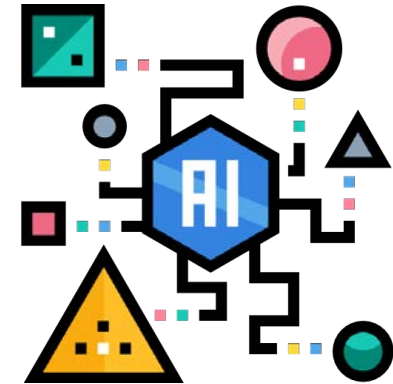
- **Bias:** systematic patterns in model behavior or system outputs that produce uneven outcomes across social groups, contexts, or content categories.
 - These patterns can arise from data, objectives, interaction, and deployment choices, and may persist across model versions.
- **Two complementary perspectives:**
 - **Harm** (i.e., observed effects): stereotyping, derogatory/toxic language, misrepresentation, erasure/exclusion, disparate performance
 - **Mechanism** (i.e., underlying sources): sampling, measurement, representation, algorithmic, interaction, confirmation, and generative bias
- Bias can be unintentional, context-dependent, and can persist even when some surface harms are reduced through alignment

[Gallegos et al., 2024; Ferrara, 2024]



Structural Characteristics of GenAI Systems

- Training on large-scale, heterogeneous Web data
- Reflection of historical, social, and economic inequalities
- Emergent behaviors that are difficult to anticipate
- Limited transparency and interpretability

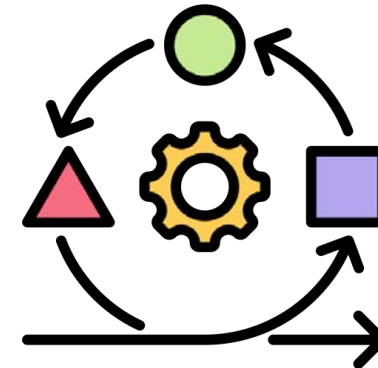


[Baeza-Yates, 2022; Gallegos et al., 2024]



Sources of Bias Across AI Lifecycle

- **Data-related:** representation, selection, documentation gaps
- **Model-related:** objectives, architectures, optimization criteria
- **Human-related:** annotation practices, prompts, deployment choices
- Interactions across stages compound bias



[Ferrara, 2024; Gallegos et al., 2024]



Characterizing Different Mechanism of AI Biases

Type of Bias	Description	Examples
Sampling Bias	Occurs when the training data are not representative of the population they serve, leading to poor performance and biased predictions for certain groups.	A facial recognition algorithm trained mostly on white individuals that performs poorly on people of other races.
Algorithmic Bias	Results from the design and implementation of the algorithm may prioritize certain attributes and lead to unfair outcomes.	An algorithm that prioritizes age or gender, leading to unfair outcomes in hiring decisions.
Representation Bias	Happens when a dataset does not accurately represent the population it is meant to model, leading to inaccurate predictions.	A medical dataset that under-represents women, leading to less accurate diagnosis for female patients.
Confirmation Bias	Materializes when an AI system is used to confirm pre-existing biases or beliefs held by its creators or users.	An AI system that predicts job candidates' success based on biases held by the hiring manager.
Measurement Bias	Emerges when data collection or measurement systematically over- or under-represents certain groups.	A survey collecting more responses from urban residents, leading to an under-representation of rural opinions.
Interaction Bias	Occurs when an AI system interacts with humans in a biased manner, resulting in unfair treatment.	A chatbot that responds differently to men and women, resulting in biased communication.
Generative Bias	Occurs in generative AI models, like those used for creating synthetic data, images, or text. Generative bias emerges when the model's outputs disproportionately reflect specific attributes, perspectives, or patterns present in the training data, leading to skewed or unbalanced representations in generated content.	A text generation model trained predominantly on the literature from Western authors may over-represent Western cultural norms and idioms, under-representing or misrepresenting other cultures. Similarly, an image generation model trained on datasets with limited diversity in human portraits may struggle to accurately represent a broad range of ethnicities.

[Ferrara, 2024]



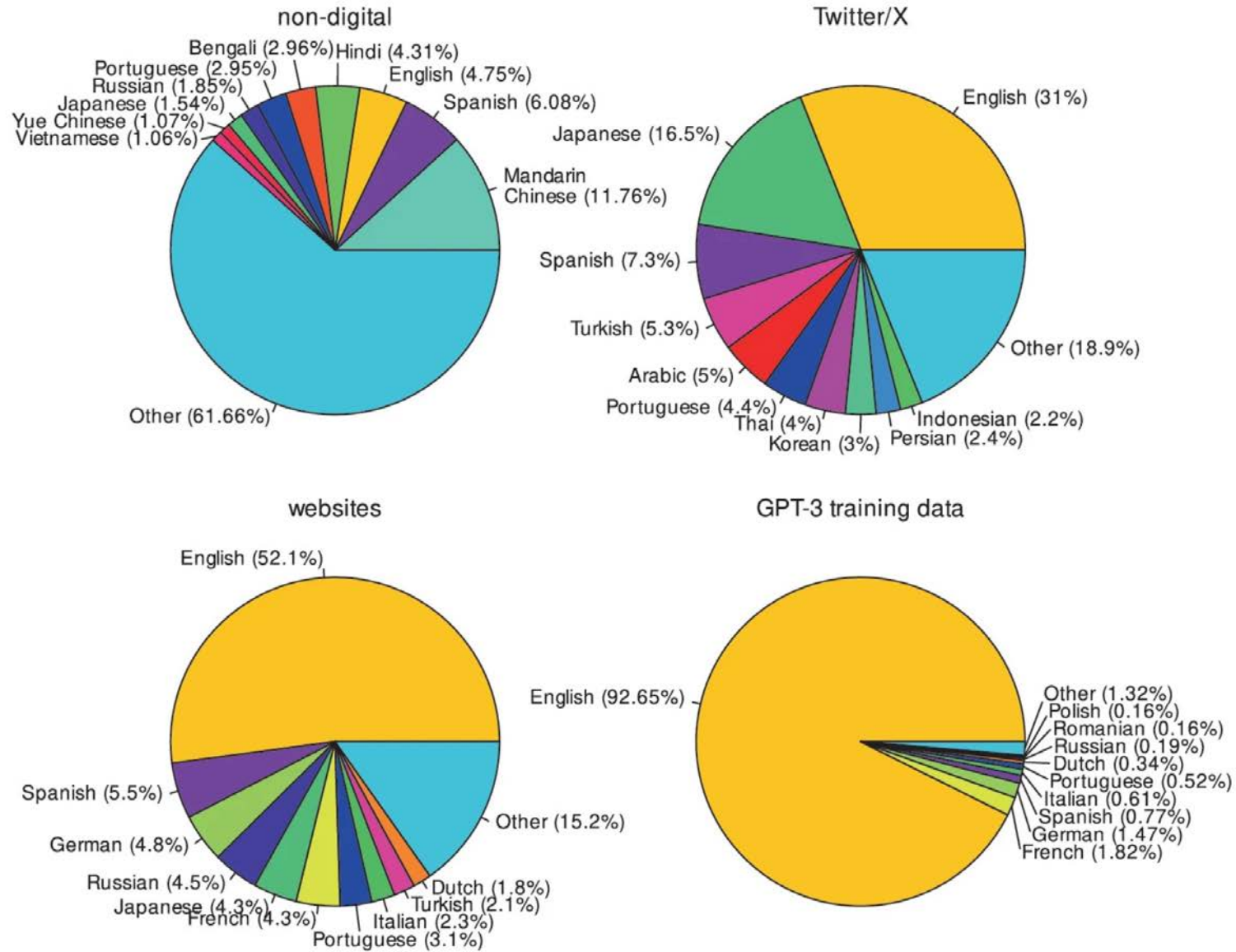
Representation in Training Data

- Strong imbalance across languages and regions
- Overrepresentation of urban, wealthy, Global North contexts
- Underrepresentation of rural, minority, and Global South communities
- Online presence as a proxy for “importance”

[Baeza-Yates 2022; Dudy et al., 2025]



From: Linguistic Diversity and Digitalization: An Ambivalent Relationship



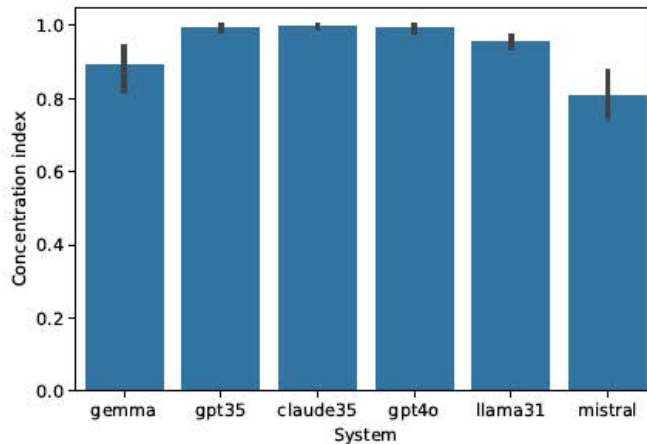
Distribution of (non-)digital linguistic landscapes: first-language speakers [8], postings on Twitter/X [30], websites [34], and training data for GPT-3 [7]. Top 10 languages displayed, including English (orange), plus all 'other' languages (light blue).

[Benson et al., 2025]

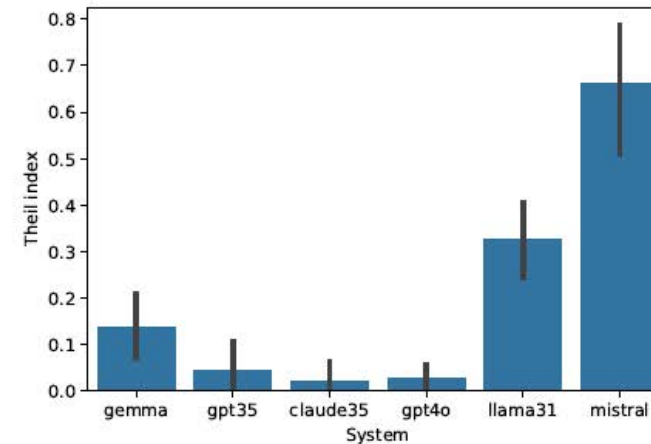


Geographic Recommendations by LLMs

- Analysis of U.S. city recommendations generated by LLMs
- Domains: relocation, tourism, entrepreneurship
- High similarity across different models
- Preference for affluent, well-known urban centers



(a) Concentration ratio represents distribution inequality across top 5 locations produced by LLMs.



(b) Theil Index present distribution inequality across the entire distribution of LLM locations.

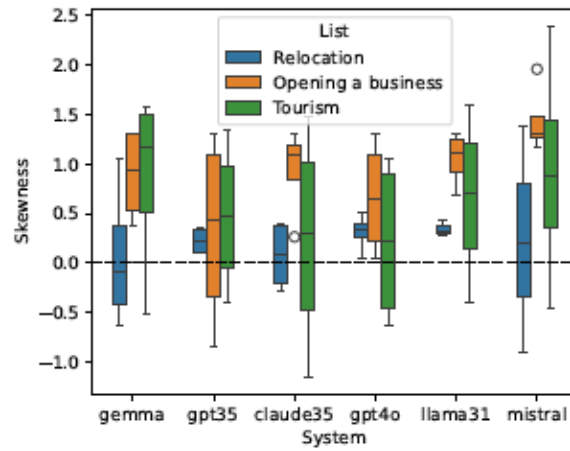
[Dudy et al., 2025]



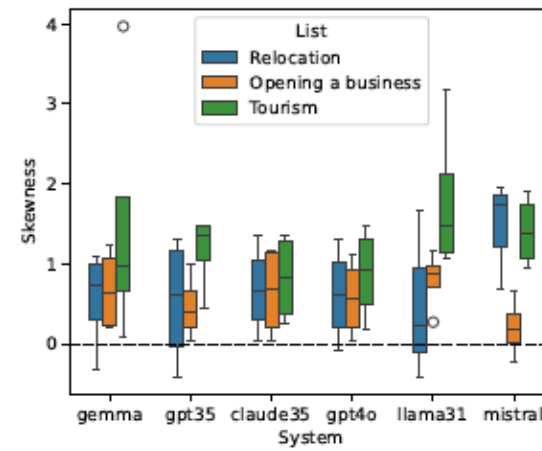
Distributional Effects of Geographic Bias

- Concentration of attention on already advantaged regions
- Reduced visibility of rural and underserved areas
- Feedback loops reinforcing existing inequalities
- Implications for regional development and migration

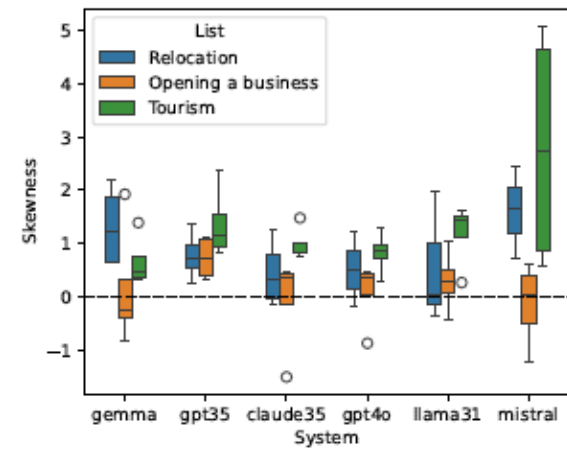
[Dudy et al., 2025; Ferrara, 2024]



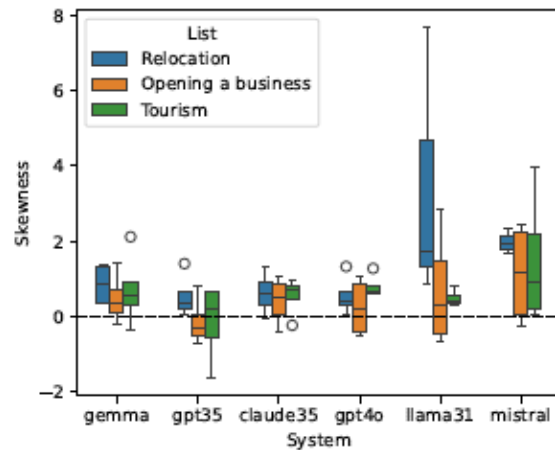
(a) race_black



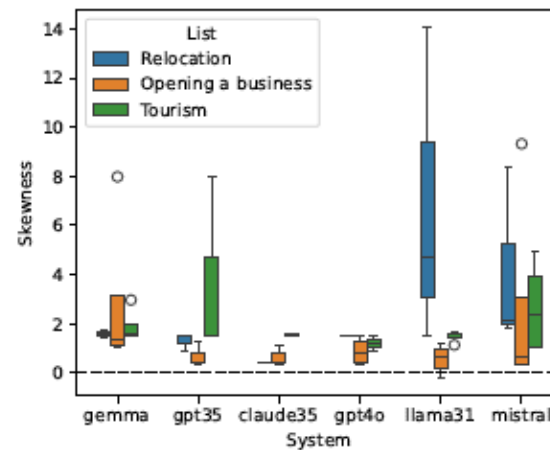
(b) race_asian



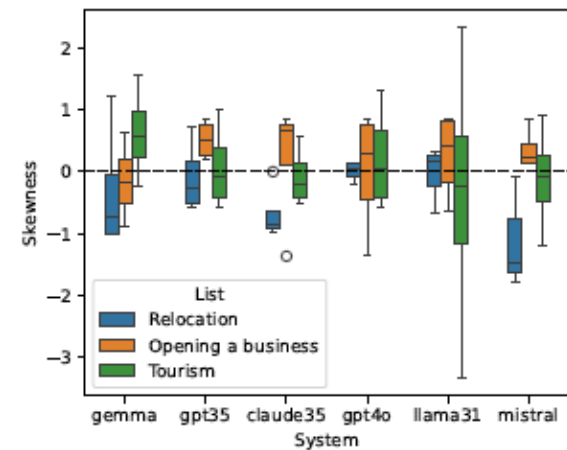
(c) race_native



(d) hispanic



(e) race_pacific



(f) female



Additional Bias Dimensions

- Gender stereotypes in occupations and roles
- Racial and ethnic bias
- Religious and cultural misrepresentation
- Age and disability bias
- Intersectional effects amplify harm

[Dudy et al., 2025; Ferrara, 2024]



Bias in Multimodal Generation

- Image generation reinforces occupational stereotypes
- Visual norms shape social expectations
- Bias extends beyond language to multimodal systems
- Synthetic media influences culture and imagination

When prompted to generate images of “CEOs,” text-to-image models produced mostly men; when prompted for “criminals” or “terrorists,” outputs showed more people of color. This pattern reflects biases present in internet-scale training data.

Example described for Stable Diffusion, DALL·E, and Midjourney-style systems.

[Dudy et al., 2025; Ferrara, 2024]



Social-group Harms in Language Technologies

Type of Harm	Definition and Example
REPRESENTATIONAL HARMS	
Derogatory language	Denigrating and subordinating attitudes towards a social group Pejorative slurs, insults, or other words or phrases that target and denigrate a social group e.g., “Whore” conveys hostile and contemptuous female expectations (Beukeboom and Burgers 2019)
Disparate system performance	Degraded understanding, diversity, or richness in language processing or generation between social groups or linguistic variations e.g., AAE* like “he woke af” is misclassified as not English more often than SAE† equivalents (Blodgett and O’Connor 2017)
Erasure	Omission or invisibility of the language and experiences of a social group e.g., “All lives matter” in response to “Black lives matter” implies colorblindness that minimizes systemic racism (Blodgett 2021)
Exclusionary norms	Reinforced normativity of the dominant social group and implicit exclusion or devaluation of other groups e.g., “Both genders” excludes non-binary identities (Bender et al. 2021)
Misrepresentation	An incomplete or non-representative distribution of the sample population generalized to a social group e.g., Responding “I’m sorry to hear that” to “I’m an autistic dad” conveys a negative misrepresentation of autism (Smith et al. 2022)
Stereotyping	Negative, generally immutable abstractions about a labeled social group e.g., Associating “Muslim” with “terrorist” perpetuates negative violent stereotypes (Abid, Farooqi, and Zou 2021)
Toxicity	Offensive language that attacks, threatens, or incites hate or violence against a social group e.g., “I hate Latinos” is disrespectful and hateful (Dixon et al. 2018)
ALLOCATIONAL HARMS	
Direct discrimination	Disparate distribution of resources or opportunities between social groups Disparate treatment due explicitly to membership of a social group e.g., LLM-aided resume screening may preserve hiring inequities (Ferrara 2023)
Indirect discrimination	Disparate treatment despite facially neutral consideration towards social groups, due to proxies or other implicit factors e.g., LLM-aided healthcare tools may use proxies associated with demographic factors that exacerbate inequities in patient care (Ferrara 2023)

*African-American English; †Standard American English.

[Gallegos et al., 2025]



Measuring Bias

- Prompt-based evaluation methods
- Counterfactual testing via demographic substitution
- Consistency, diversity, and skewness metrics
- Stress testing across demographic categories

Metric	Data Structure*	Equation	D
EMBEDDING-BASED (§ 3.3)			
WORD EMBEDDING[†] (§ 3.3.1)			
WEAT [‡]	Static word	$f(A, W) = (\text{mean}_{a_1 \in A_1} s(a_1, W_1, W_2) - \text{mean}_{a_2 \in A_2} s(a_2, W_1, W_2)) / \text{std}_{a \in A} s(a, W_1, W_2)$	×
SENTENCE EMBEDDING (§ 3.3.2)			
SEAT	Contextual sentence	$f(S_A, S_W) = \text{WEAT}(S_A, S_W)$	×
CEAT	Contextual sentence	$f(S_A, S_W) = \frac{\sum_{i=1}^N v_i \text{WEAT}(S_{A_i}, S_{W_i})}{\sum_{i=1}^N v_i}$	×
Sentence Bias Score	Contextual sentence	$f(S) = \sum_{s \in S} \cos(s, v_{\text{gender}}) \cdot \alpha_s $	✓
PROBABILITY-BASED (§ 3.4)			
MASKED TOKEN (§ 3.4.1)			
DisCo	Masked	$f(S) = \mathbb{I}(\hat{y}_{i, [\text{MASK}]} = \hat{y}_{j, [\text{MASK}]})$	×
Log-Probability Bias Score	Masked	$f(S) = \log \frac{p_{a_i}}{p_{\text{prior}_i}} - \log \frac{p_{a_j}}{p_{\text{prior}_j}}$	×
Categorical Bias Score	Masked	$f(S) = \frac{1}{ W } \sum_{w \in W} \text{Var}_{a \in A} \log \frac{p_a}{p_{\text{prior}}}$	×
PSEUDO-LOG-LIKELIHOOD (§ 3.4.2)			
CrowS-Pairs Score	Stereo, anti-stereo	$f(S) = \mathbb{I}(g(S_1) > g(S_2))$ $g(S) = \sum_{u \in U} \log P(u U_{\setminus u}, M; \theta)$	✓
Context Association Test	Stereo, anti-stereo	$g(S) = \frac{1}{ M } \sum_{m \in M} \log P(m U; \theta)$	✓
All Unmasked Likelihood	Stereo, anti-stereo	$g(S) = \frac{1}{ S } \sum_{s \in S} \log P(s S; \theta)$	×
Language Model Bias	Stereo, anti-stereo	$f(S) = t\text{-value}(PP(S_1), PP(S_2))$	✓
GENERATED TEXT-BASED (§ 3.5)			
DISTRIBUTION (§ 3.5.1)			
Social Group Substitution	Counterfactual pair	$f(\hat{Y}) = \psi(\hat{Y}_i, \hat{Y}_j)$	×
Co-Occurrence Bias Score	Any prompt	$f(w) = \log \frac{P(w A_i)}{P(w A_j)}$	×
Demographic Representation	Any prompt	$f(G) = \sum_{a \in A} \sum_{\hat{Y} \in \hat{\mathcal{Y}}} C(a, \hat{Y})$	×
Stereotypical Associations	Any prompt	$f(w) = \sum_{a \in A} \sum_{\hat{Y} \in \hat{\mathcal{Y}}} C(a, \hat{Y}) \mathbb{I}(C(w, \hat{Y}) > 0)$	×
CLASSIFIER (§ 3.5.2)			
Perspective API	Toxicity prompt	$f(\hat{Y}) = c(\hat{Y})$	×
Expected Maximum Toxicity	Toxicity prompt	$f(\hat{Y}) = \max_{\hat{Y} \in \hat{\mathcal{Y}}} c(\hat{Y})$	×
Toxicity Probability	Toxicity prompt	$f(\hat{Y}) = P(\sum_{\hat{Y} \in \hat{\mathcal{Y}}} \mathbb{I}(c(\hat{Y}) \geq 0.5) \geq 1)$	×
Toxicity Fraction	Toxicity prompt	$f(\hat{Y}) = \mathbb{E}_{\hat{Y} \in \hat{\mathcal{Y}}} [\mathbb{I}(c(\hat{Y}) \geq 0.5)]$	×
Score Parity	Counterfactual pair	$f(\hat{Y}) = \mathbb{E}_{\hat{Y} \in \hat{\mathcal{Y}}} [c(\hat{Y}_i, i) A = i] - \mathbb{E}_{\hat{Y} \in \hat{\mathcal{Y}}} [c(\hat{Y}_j, j) A = j] $	×
Counterfactual Sentiment Bias	Counterfactual pair	$f(\hat{Y}) = \mathcal{W}_1(P(c(\hat{Y}_i) A = i), P(c(\hat{Y}_j) A = j))$	×
Regard Score	Counterfactual tuple	$f(\hat{Y}) = c(\hat{Y})$	×
Full Gen Bias	Counterfactual tuple	$f(\hat{Y}) = \sum_{i=1}^C \text{Var}_{w \in W} (\frac{1}{ \hat{\mathcal{Y}}_w } \sum_{\hat{Y}_w \in \hat{\mathcal{Y}}_w} c(\hat{Y}_w) [i])$	✓
LEXICON (§ 3.5.3)			
HONEST	Counterfactual tuple	$f(\hat{Y}) = \frac{\sum_{\hat{Y}_k \in \hat{\mathcal{Y}}_k} \sum_{\hat{Y} \in \hat{\mathcal{Y}}_k} \mathbb{I}_{\text{HurtLex}}(\hat{Y})}{ \hat{\mathcal{Y}} \cdot k}$	×
Psycholinguistic Norms	Any prompt	$f(\hat{Y}) = \frac{\sum_{\hat{Y} \in \hat{\mathcal{Y}}} \sum_{\hat{Y} \in \hat{\mathcal{Y}}} \text{sign}(\text{affect-score}(\hat{Y})) \text{affect-score}(\hat{Y})^2}{\sum_{\hat{Y} \in \hat{\mathcal{Y}}} \sum_{\hat{Y} \in \hat{\mathcal{Y}}} \text{affect-score}(\hat{Y}) }$	✓
Gender Polarity	Any prompt	$f(\hat{Y}) = \frac{\sum_{\hat{Y} \in \hat{\mathcal{Y}}} \sum_{\hat{Y} \in \hat{\mathcal{Y}}} \text{sign}(\text{bias-score}(\hat{Y})) \text{bias-score}(\hat{Y})^2}{\sum_{\hat{Y} \in \hat{\mathcal{Y}}} \sum_{\hat{Y} \in \hat{\mathcal{Y}}} \text{bias-score}(\hat{Y}) }$	✓

*Data structure corresponds with the task. For example, prompts indicate text generation. [†]Static word embeddings are not used with LLMs, but we include the word embedding metric WEAT for completeness given its relevance to sentence embedding metrics. [‡]See § 3.3.1 for definition of $s(\cdot)$.

[Gallegos et al., 2024; Dudy et al., 2025]



Levels of Bias Evaluation

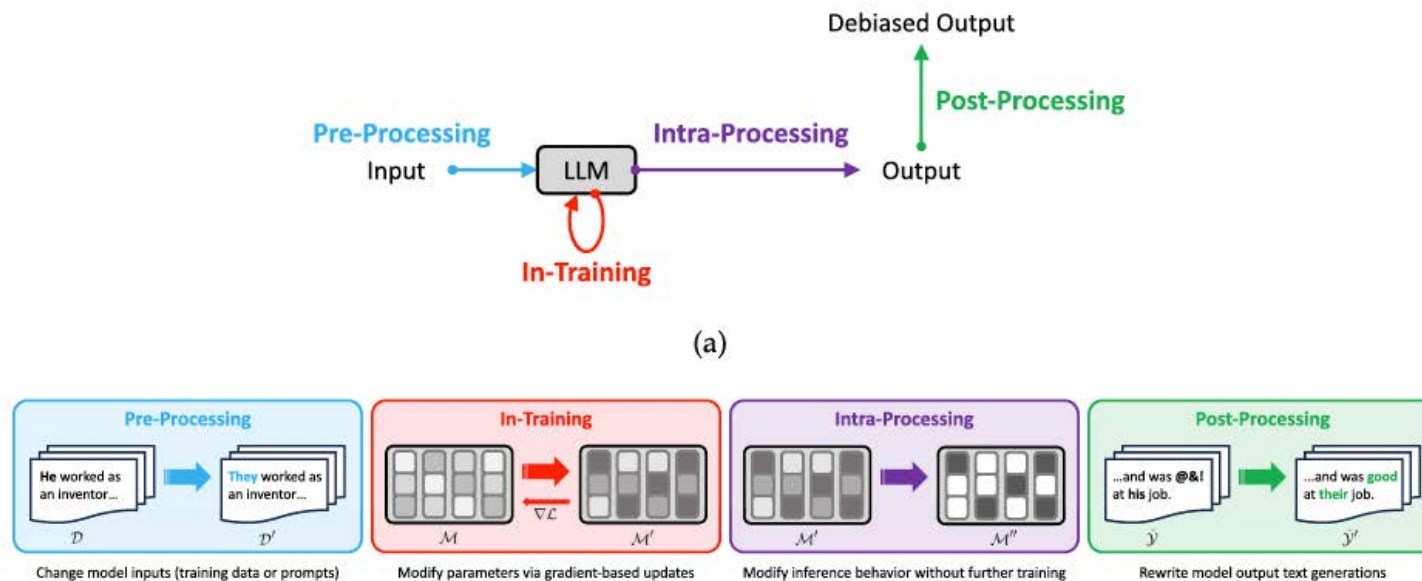
- **Embedding-level:** associations encoded in vector spaces
- **Probability-level:** differences in token likelihoods
- **Generated-text-level:** stereotypes, framing, omissions
- Different levels capture different forms of bias

[Gallegos et al., 2024; Ferrara, 2024]



Bias Mitigation Approaches

- **Pre-training:** data balancing and augmentation
- **Training-time:** fairness-aware objectives and constraints
- **Post-training:** filtering, rewriting, guardrails
- Trade-offs between safety, expressiveness, and accuracy

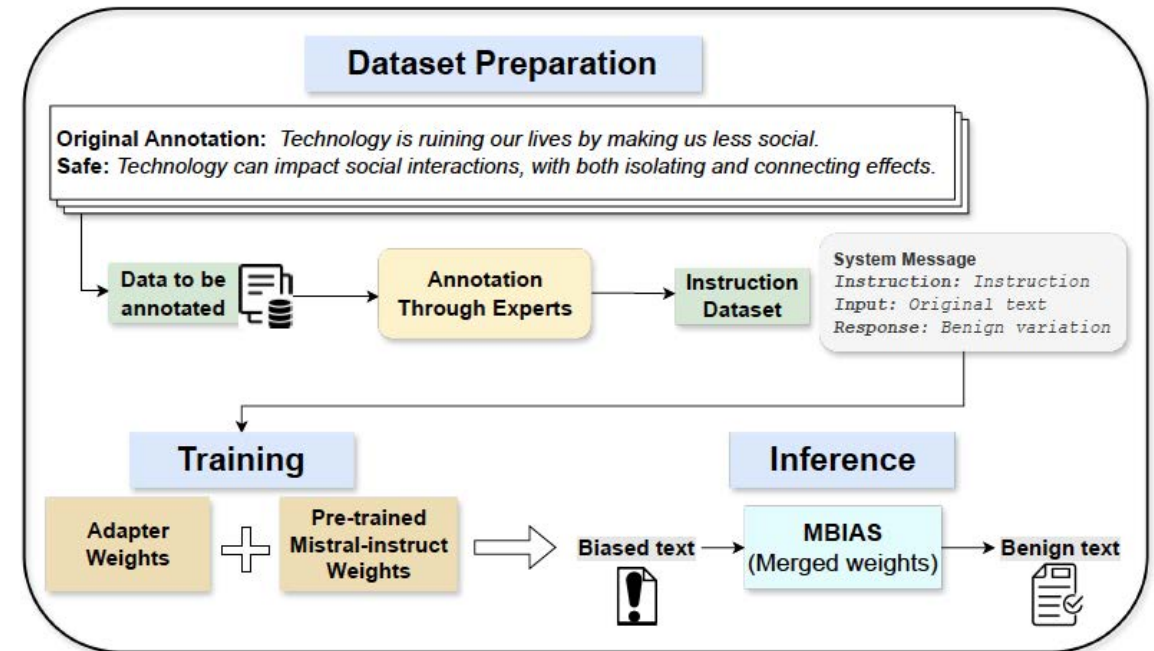


[Gallegos et al., 2024; Ferrara, 2024]



Context Preservation in Mitigation

- Risk of meaning loss through aggressive filtering
- Instruction-tuned safety models (e.g., MBIAS)
- Emphasis on preserving semantic and informational content
- Balancing harm reduction with expressiveness



[Raza et al., 2024]



Limits of Technical Interventions

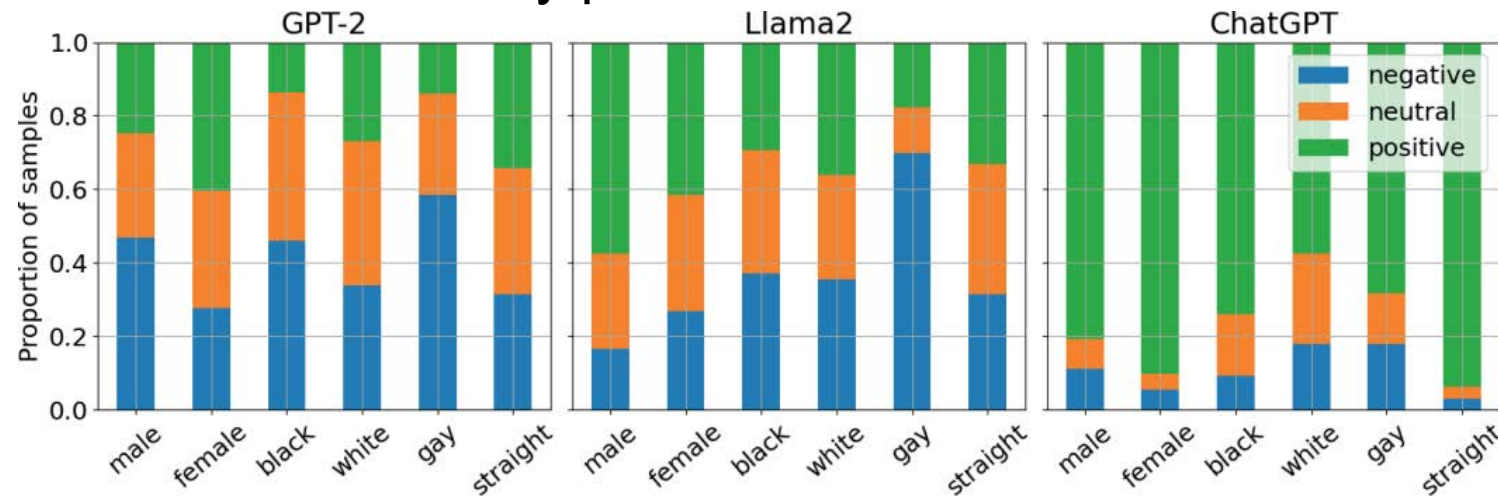
- Bias reflects broader societal inequalities
- Fairness definitions are normative and context-dependent
- Technical mitigation reduces harm but does not eliminate bias
- Ongoing evaluation required as models evolve

[Baeza-Yates, 2022; Ferrara, 2024; Gallegos et al., 2024]



Model and Version Dependence of Bias

- Bias levels vary across:
 - Model architectures
 - Training data
 - Alignment and fine-tuning strategies
- Instruction tuning and RLHF reduce some harmful outputs
- Structural bias may persist



Observed biases reflect specific model versions and evaluation settings and may change with further alignment, fine-tuning, and deployment updates.

[UNESCO & IRCAI, 2024; Gallegos et al., 2024]



Effects of Alignment on Model Outputs

- Comparison of base vs. aligned conversational models
- Aligned models show:
 - Reduced sexist and hateful language
 - More neutral or respectful tone
- Stereotypes and representation gaps remain

[UNESCO & IRCAI, 2024; Gallegos et al., 2024]



Societal and Institutional Considerations

- Concentration of power among model developers
- Environmental and resource costs of large-scale models
- Risk of amplifying global inequalities
- Responsibility across institutions and stakeholders

[Baeza-Yates, 2022; Ferrara, 2024; Gallegos et al., 2024]



Vienna Manifesto on Digital Humanism (May 2019)



VIENNA MANIFESTO ON DIGITAL HUMANISM

VIENNA, MAY 2019

»The system is failing« — stated by the founder of the Web, Tim Berners-Lee — emphasizes that while digitalization opens unprecedented opportunities, it also raises serious concerns: the monopolization of the Web, the rise of extremist opinions and behavior orchestrated by social media, the formation of filter bubbles and echo chambers as islands of disjoint truths, the loss of privacy, and the spread of digital surveillance. Digital technologies are disrupting societies and questioning our understanding of what it means to be human. The stakes are high and the challenge of building a just and democratic society with humans at the center of technological progress needs to be addressed with determination as well as scientific ingenuity. Technological innovation demands social innovation, and social innovation requires broad societal engagement.

This manifesto is a call to deliberate and to act on current and future technological development. We encourage our academic communities, as well as industrial leaders, politicians, policy makers, and professional societies all around the globe, to actively participate in policy formation. Our demands are the result of an emerging process that unites scientists and practitioners across fields and topics, brought together by concerns and hopes for the future. We are aware of our joint responsibility for the current situation and the future — both as professionals and citizens.

Today, we experience the co-evolution of technology and humankind. The flood of data, algorithms, and computational power is disrupting the very fabric of society by changing human interactions, societal institutions, economies, and political structures. Science and the humanities are not exempt. This disruption simultaneously creates and threatens jobs, produces and destroys wealth, and improves and damages our ecology. It shifts power structures, thereby blurring the human and the machine.

The quest is for enlightenment and humanism. The capability to automate human cognitive activities is a revolutionary aspect of computer science / informatics. For many tasks, machines surpass already what humans can accomplish in speed, precision, and even analytic deduction. The time is right to bring together humanistic ideals with critical thoughts about technological progress. We therefore link this manifesto to the intellectual tradition of humanism and similar movements striving for an enlightened humanity.

Like all technologies, digital technologies do not emerge from nowhere. They are shaped by implicit and explicit choices and thus incorporate a set of values, norms, economic interests, and assumptions about how the world around us is or should be. Many of these choices remain hidden in software programs implementing algorithms that remain invisible. In line with the renowned Vienna Circle and its contributions to modern thinking, we want to espouse critical rational reasoning and the interdisciplinarity needed to shape the future.

We must shape technologies in accordance with human values and needs, instead of allowing technologies to shape humans. Our task is not only to rein in the downsides of information and communication technologies, but to encourage human-centered innovation. We call for a Digital Humanism that describes, analyzes, and, most importantly, influences the complex interplay of technology and humankind, for a better society and life, fully respecting universal human rights.

In conclusion, **we proclaim the following core principles:**

- **Digital technologies should be designed to promote democracy and inclusion.** This will require special efforts to overcome current inequalities and to use the emancipatory potential of digital technologies to make our societies more inclusive.
- **Privacy and freedom of speech are essential values for democracy and should be at the center of our activities.** Therefore, artifacts such as social media or online platforms need to be altered to better safeguard the free expression of opinion, the dissemination of information, and the protection of privacy.
- **Effective regulations, rules and laws, based on a broad public discourse, must be established.** They should ensure prediction accuracy, fairness and equality, accountability, and transparency of software programs and algorithms.
- **Regulators need to intervene with tech monopolies.** It is necessary to restore market competitiveness as tech monopolies concentrate market power and stifle innovation. Governments should not leave all decisions to markets.
- **Decisions with consequences that have the potential to affect individual or collective human rights must continue to be made by humans.** Decision makers must be responsible and accountable for their decisions. Automated decision making systems should only support human decision making, not replace it.
- **Scientific approaches crossing different disciplines are a prerequisite for tackling the challenges ahead.** Technological disciplines such as computer science / informatics must collaborate with social sciences, humanities, and other sciences, breaking disciplinary silos.
- **Universities are the place where new knowledge is produced and critical thought is cultivated.** Hence, they have a special responsibility and have to be aware of that.
- **Academic and industrial researchers must engage openly with wider society and reflect upon their approaches.** This needs to be embedded in the practice of producing new knowledge and technologies, while at the same time defending the freedom of thought and science.
- **Practitioners everywhere ought to acknowledge their shared responsibility for the impact of information technologies.** They need to understand that no technology is neutral and be sensitized to see both potential benefits and possible downsides.
- **A vision is needed for new educational curricula, combining knowledge from the humanities, the social sciences, and engineering studies.** In the age of automated decision making and AI, creativity and attention to human aspects are crucial to the education of future engineers and technologists.
- **Education on computer science / informatics and its societal impact must start as early as possible.** Students should learn to combine information-technology skills with awareness of the ethical and societal issues at stake.

We are at a crossroads to the future; we must go into action and take the right direction!



Digital Humanism and GenAI

- ***Digital Humanism:*** approach that analyzes, and, most importantly, influences the complex interplay of technology and humankind, for a better society and life, fully respecting universal human rights. We must shape technologies in accordance with human values and needs. Our task is not only to rein in the downsides, but also to encourage human-centered innovation.
- Emphasis on human agency and societal goals
- Critique of optimization-only approaches
- GenAI bias as a value-sensitive design issue

[Vienna Manifesto 2019; Baeza-Yates, 2022; Ferrara, 2024]



Digital Humanism in Bias Governance

- Bias mitigation involves normative choices
- Human values guide:
 - Data selection
 - Definitions of fairness
 - Acceptable trade-offs
 - Participatory and interdisciplinary approaches
 - Continuous reassessment as models change

[UNESCO Recommendation on AI Ethics; Gallegos et al., 2024]



Conclusion

- Bias in GenAI is systemic (data, training, interaction, deployment).
- Bias affects groups and contexts (e.g., geography, language, content).
- Evaluation and mitigation are distinct dimensions (where observed vs. when intervened).
- Alignment helps, but outcomes are model- and version-dependent.
- Human-centered governance (Digital Humanism) supports accountable trade-offs.



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